Data Driven Control Integration Week

Group 8 (MPC)

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Friday, November 6, 2026

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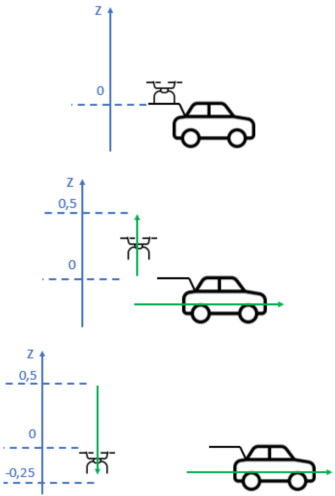
Project Objectives

Available Resources:

- Dataset of vertical acceleration.
- MATLAB/Simulink drone simulator.

Objectives:

- Validate the dataset.
- Identify the model.
- Develop a control law.



Dataset Validation

Cleaning the data

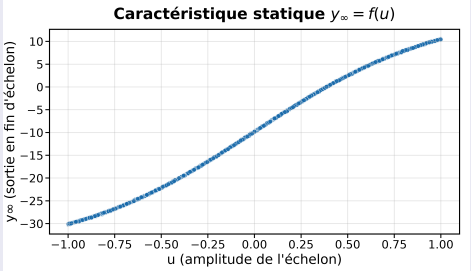
Remove lines that includes:

- Frozen sensor for 2+ seconds
- 5+ diverging values

Normalisation

- Directly in the training

Static characteristic



$y_\infty = f(u)$ after eliminating corrupted runs.

System Modeling

Linear Model

State Function

First-order lag driven by $N(u)$:

$$\begin{cases} \dot{x}(t) = -\dfrac{x(t)}{\tau} + \dfrac{N(u(t))}{\tau} \\ a_z(t) = x(t) \end{cases}$$

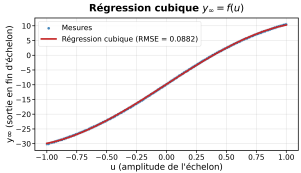
Parameter Identification

- **Static characteristic** $N(u)$: 3rd-order polynomial identified on steady-state regimes:

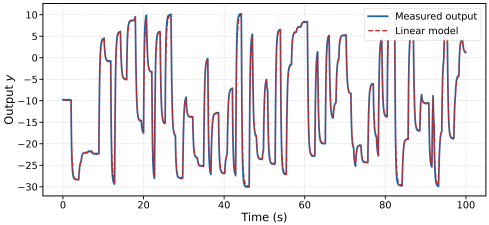
$$a_z = N(u) = a_3 u^3 + a_2 u^2 + a_1 u + a_0$$

- **Tau identification**

$a_3 \approx -6.05$, $a_2 \approx 1.1\times10^{-3}$, $a_1 \approx 26.1$, $a_0 \approx -9.81$, $\tau \approx 0.22$ s



Linear Model Validation

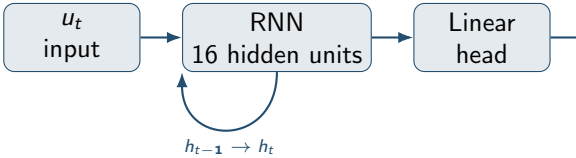


First RNN

Sequence-to-sequence architecture

Parameters

- One scalar input u_t and one scalar output $\hat{y}_t$
- Hidden state dimension: 16
- Early stopping to avoid overfitting



Forward equations

$$h_t = \tanh(W_{ih}u_t + W_{hh}h_{t-1} + b)$$

$$\hat{y}_t = W_{hy}h_t + b_y$$

Least-squares criterion

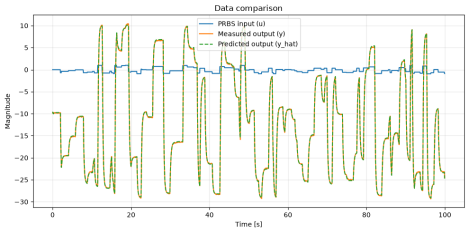
$$\sum^{N}$$

First RNN

Results

Numerical results

- Validation loss: 3.93×10^{-4}
- Test loss: 4.01×10^{-4}
- Training completed at epoch 100
- Only 321 trainable parameters



Z-axis control

Designing the reference trajectory

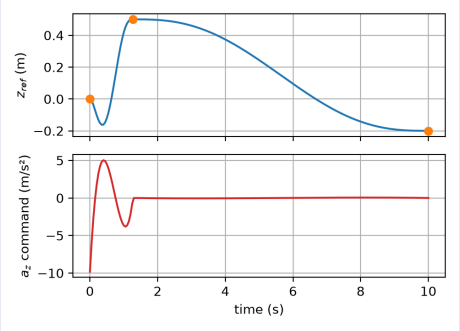
Principle

- **Input**: waypoints z_i and passage times t_i
- One **quintic polynomial** per segment $[t_i, t_{i+1}]$, imposing z, $\dot{z}$ and $\ddot{z}$ at both ends (jerk free)
- Null speed at waypoints, null acceleration on arrival $\Rightarrow$ $\ddot{z}$ continuous
- **Output**: $z_{ref}(t)$ and the acceleration command $\ddot{z}_{ref}(t)$

Results

Smooth analytic reference; overshoot beyond waypoints removed by shortening segments

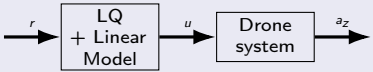
Example: $z = 0, 0.5, -0.2$ m at $t = 0, 5, 10$ s



LQ control

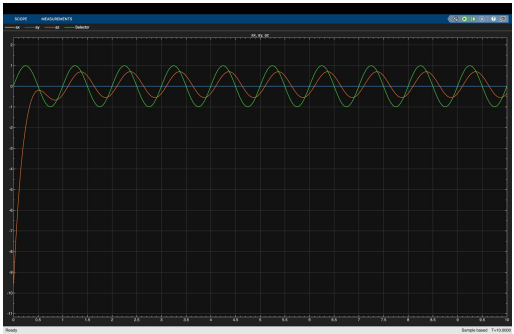
Linear-quadratic control design

Control principle



Performance

- Global trajectory following
- Wrong amplitude + phase shift
- Missing prediction



MPC control

Model predictive control design

Control principle

Replace this text with the main elements of the MPC controller:

- Describe the prediction model.
- Explain the prediction and control horizons.
- Present the cost function and constraints.

Performance

Add here the main performance result or metric.

Insert MPC control figure here

Appendix: System Modeling

Linear Model: Least Square Method

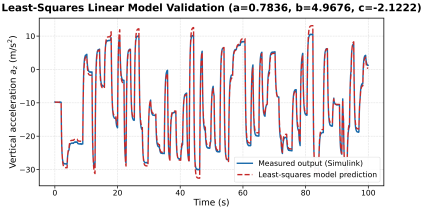
Model equations

$$y_k(t+1) = a\,y_k(t) + b\,u_k(t) + c$$
$$= \varphi_k^{\mathsf{T}}\theta.$$

$$\varphi_k^{\mathsf{T}} = \begin{bmatrix} y_k(t) & u_k(t) & 1 \end{bmatrix} \qquad \theta = \begin{bmatrix} a & b & c \end{bmatrix}^{\mathsf{T}}.$$

The least-squares estimate is

$$\theta^{\star} = \left(\Phi^{\mathsf{T}}\Phi\right)^{-1}\Phi^{\mathsf{T}}Y.$$



Appendix: System Modeling

Linear Model with Constant Steady-State Gain

State Function

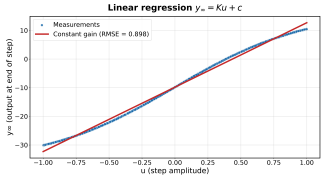
First-order lag driven by a constant gain K and offset c: $\dot{v}(t) = \frac{1}{\tau}\big(Ku(t) - v(t)\big)$, i.e. $\tau\,\dot{v} + v = Ku + c$.

Parameter Identification

- **Static gain** K: affine least-squares fit on steady-state regimes:

$$v(t) = K\,u(t) + c, \qquad K \approx 22.5,\ c \approx -9.8$$

- **Tau identification**



Linear Model Validation

